Digital Radiography: An Overview

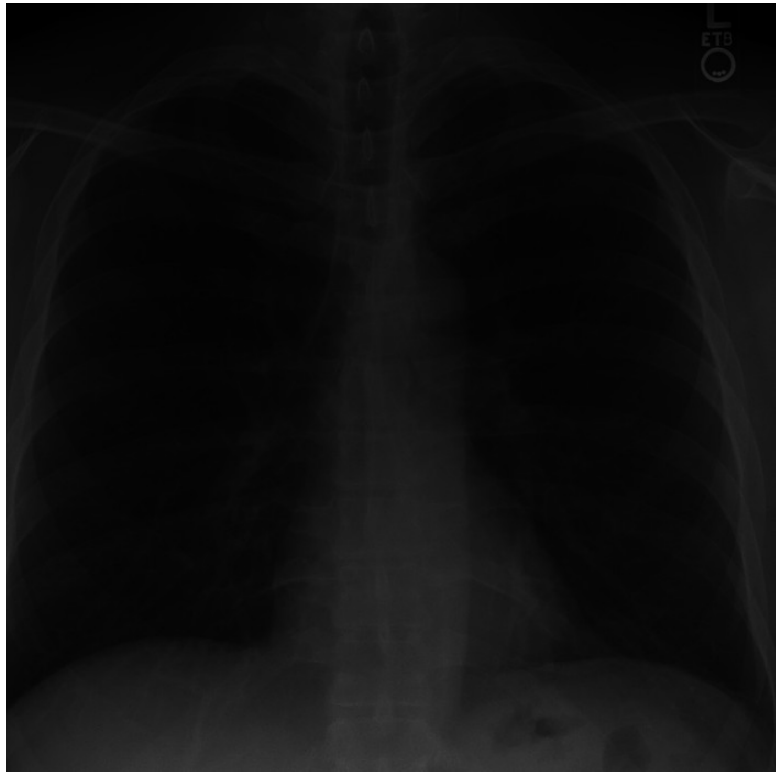
From film to pixels to PACS

Diagnostic Imaging · Homework 1, Topic 1

Group: A · B · C · D (replace with names)

Primary source: Seeram E. (2019), Digital Radiography, 2nd ed., Springer, Ch. 1

Same mistake, two outcomes

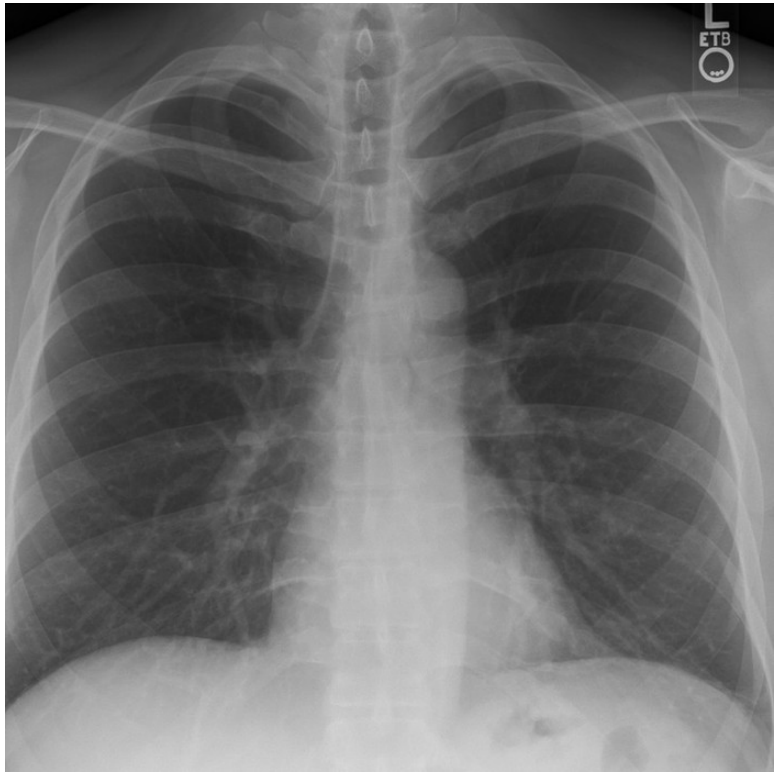


Film

almost black: repeat needed

Same chest.
Same 4× mAs.

- Film
- Detectors
- Dose
- PACS



Digital (DR)

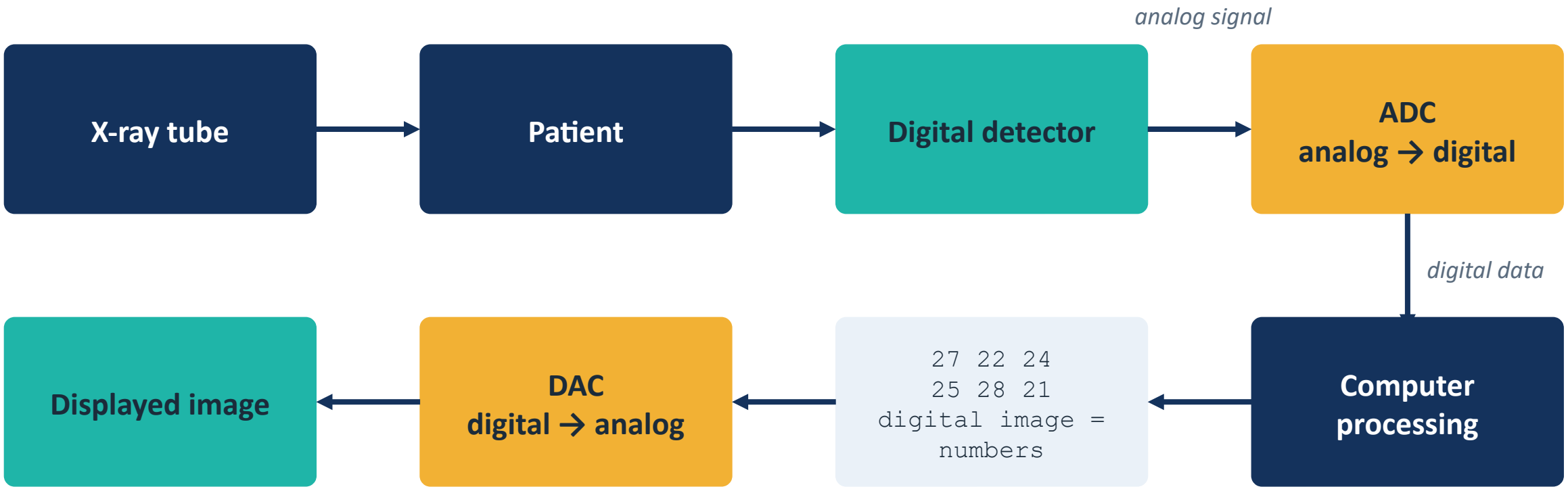
looks normal

Which patient received too much radiation?

Images: simulated by the group from a CC0 chest PA radiograph (Wikimedia Commons, Stillwaterising); film via an H&D-curve model, cf. Seeram (2019) Figs 1.3-1.4. Noise/exposure exaggerated.

What is digital radiography?

Digital detectors turn X-ray exposure into numbers, and give usable images over a wider range of exposures than film.



Radiologists never see the numbers, only the displayed image.

Redrawn from Seeram (2019) Fig 1.1 and Fig 1.5; definition after AAPM Report 116, quoted in Seeram §1.2.

How film works, and its S-curve

X-ray exposure

↓

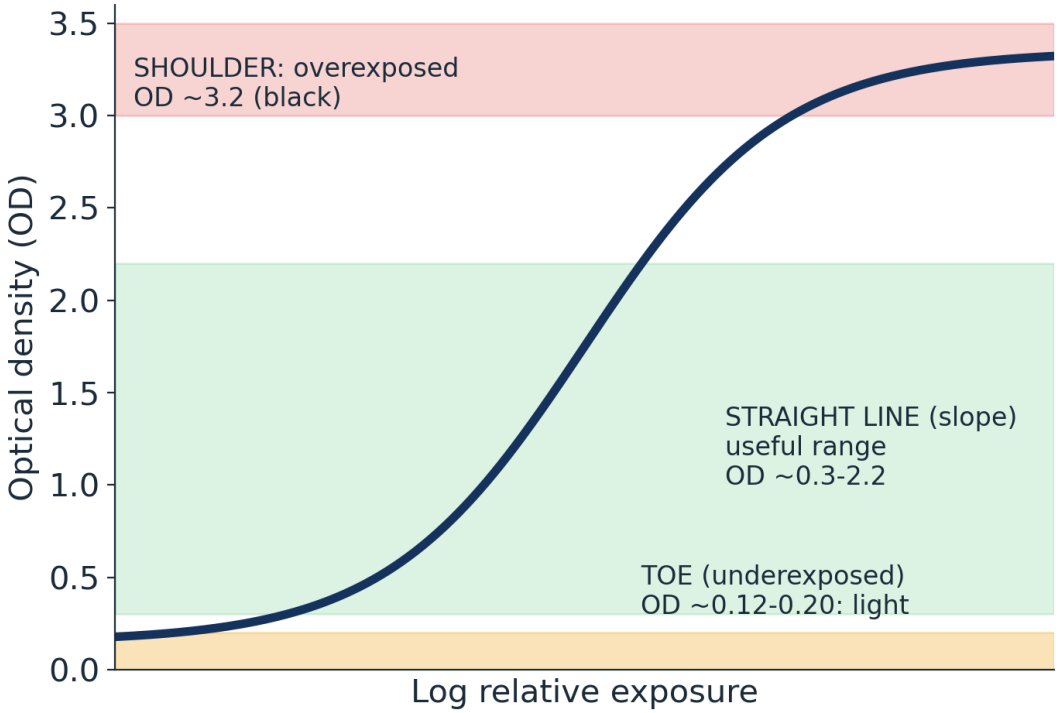
Latent image

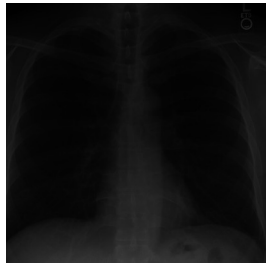
↓

Chemical processing

↓

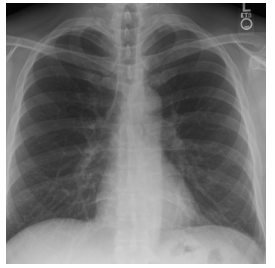
Viewbox





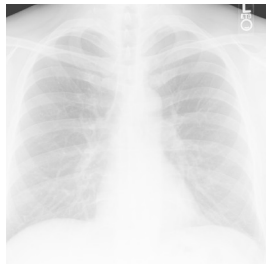
Shoulder

overexposed



Straight line

useful



Toe

underexposed

$OD = \log_{10}(I_0 / I_t)$ · Film = detector + display + archive in one

Redrawn from Seeram (2019) Figs 1.2 and 1.4 (OD values from §1.3.2). Thumbnails: simulated by the group from a CC0 chest radiograph.

Exposure factors: what the technologist controls

	kVp (tube voltage)	mAs (tube current × time)
Controls	Beam quality: penetrating power	Beam quantity: number of photons
Main image effect	Contrast: lower kVp = higher contrast	Noise: low mAs = grainy (quantum mottle)
Patient dose	≈ kVp^2 at fixed mAs	Proportional: 2× mAs = 2× dose

AEC stops the exposure at a preset detector level (film: optical density · digital: signal-to-noise)

Seeram (2019) §1.3.3-1.3.6. Dose-vs-kVp relation is an approximation at fixed mAs.

Why film had to go

1

Wrong exposure = repeat

Too light or too dark film is unusable. A repeat means extra dose.

2

The display is fixed

Brightness and contrast are set at exposure. A change needs a new exposure.

3

The archive is physical

Envelopes, large rooms, manual retrieval. One copy in one place.

Film’s one strength: spatial resolution ≈ 5–15 lp/mm

Seeram (2019) §1.3.7.

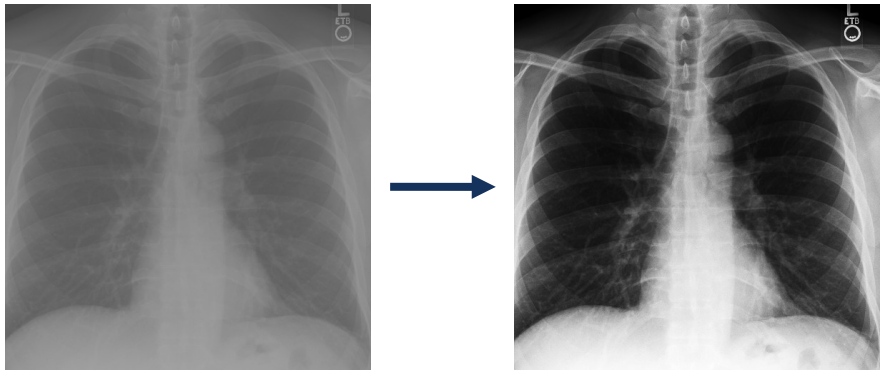
Inside a digital radiography system

1	2	3	4	5	6
Acquisition	**ADC + computer**	**Display + post-processing**	**Storage**	**Communications**	**Management**
digital detector	binary processing	DAC, monitor, grayscale mapping	RAID, tape, optical disk	LAN inside, WAN outside	PACS, RIS, HIS

Computers store everything as bits

EUCLID in binary:

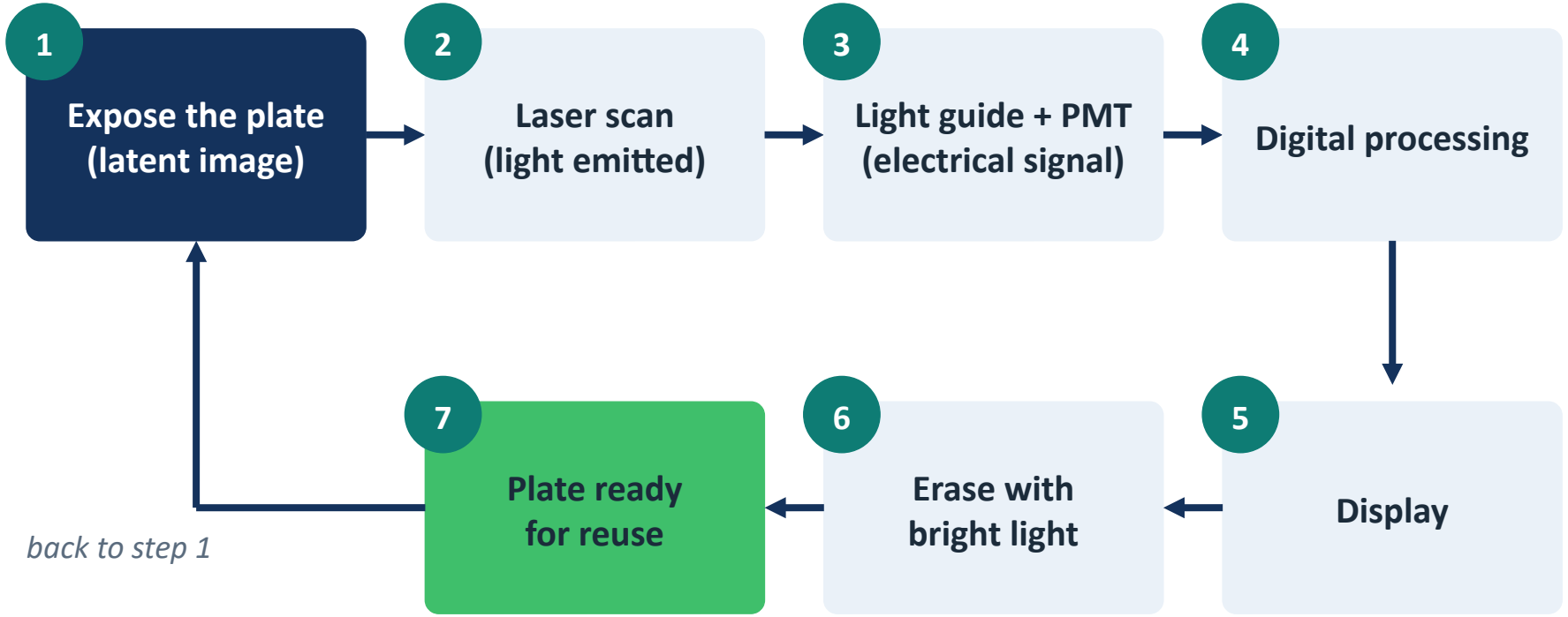
```
01000101 01010101 01000011
01001100 01001001 01000100
```



Grayscale mapping: low → high contrast

Redrawn from Seeram (2019) Fig 1.5 and §1.4; grayscale mapping after Fig 1.6 (same CC0 chest radiograph, processed by the group).

Computed radiography (CR): the reusable plate



Phosphor
BaFBr:Eu^{2+} on an imaging plate (IP)

Introduced
Fuji, 1983

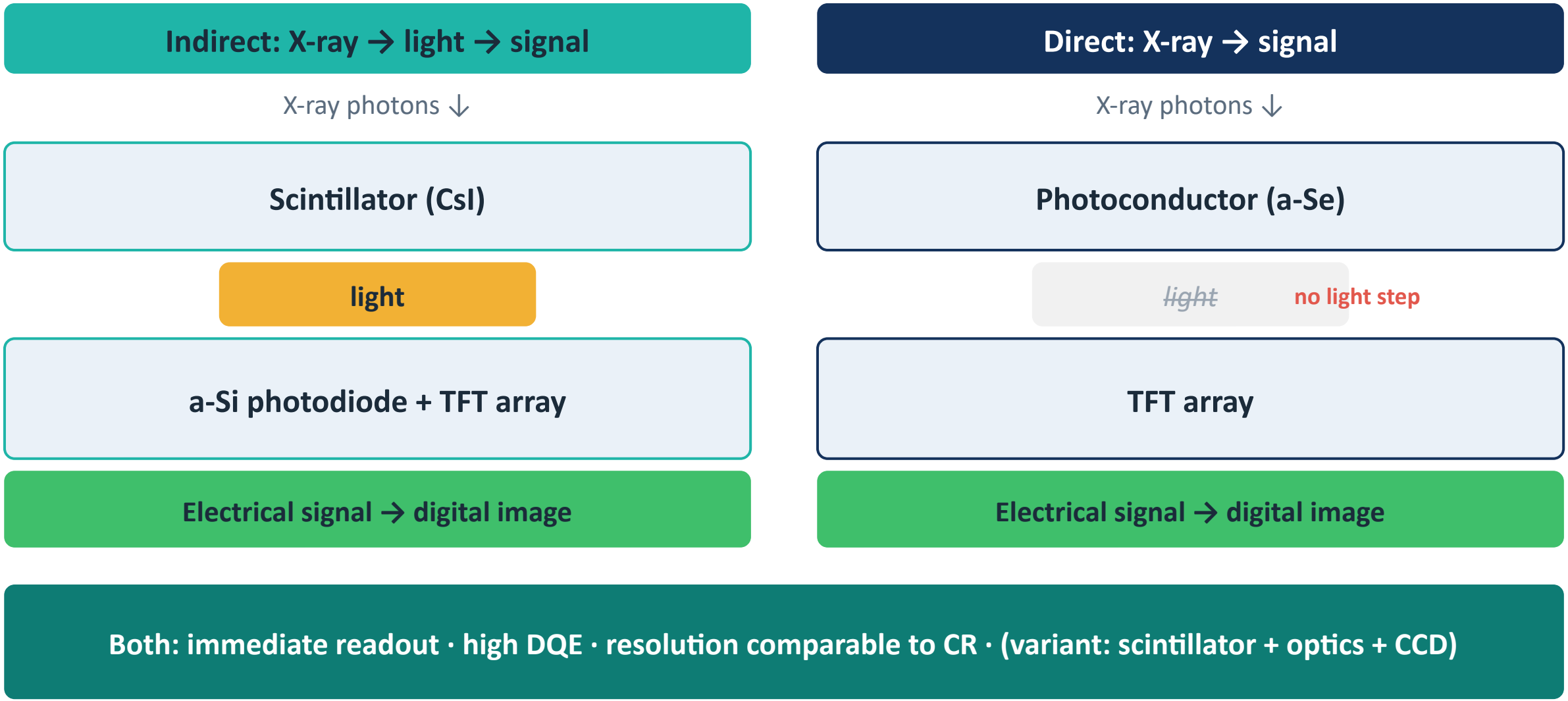
Spatial resolution
≈ 3–5 lp/mm

Dynamic range
≈ 10,000 : 1 (far wider than film)

Latent image = trapped electrons; the laser frees them as light

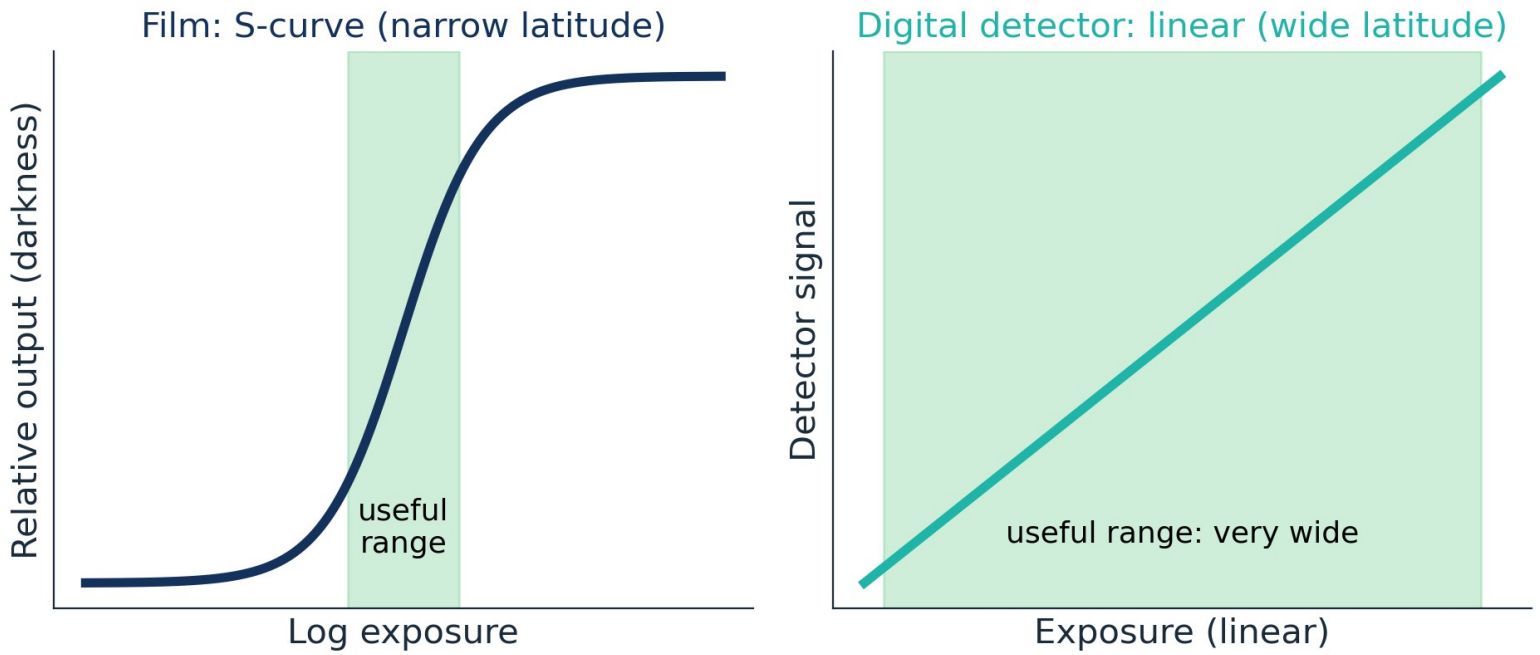
Redrawn from Seeram (2019) Fig 1.8 and §1.6.1. Laser/emission details: Storage Phosphors for Medical Imaging, Materials 2011 (PMC5448636).

Flat-panel detectors: indirect vs direct



Redrawn from Seeram (2019) Figs 1.9 and 1.10, §1.6.2. Terms (a-Si photodiode + TFT): see sources.md.

Straight line vs S-curve: exposure latitude



Too little exposure
→ grain (noise) shows

Too much exposure
→ looks fine, dose is hidden

Digital dynamic range ≈ 10,000 : 1; film's useful latitude is far narrower.

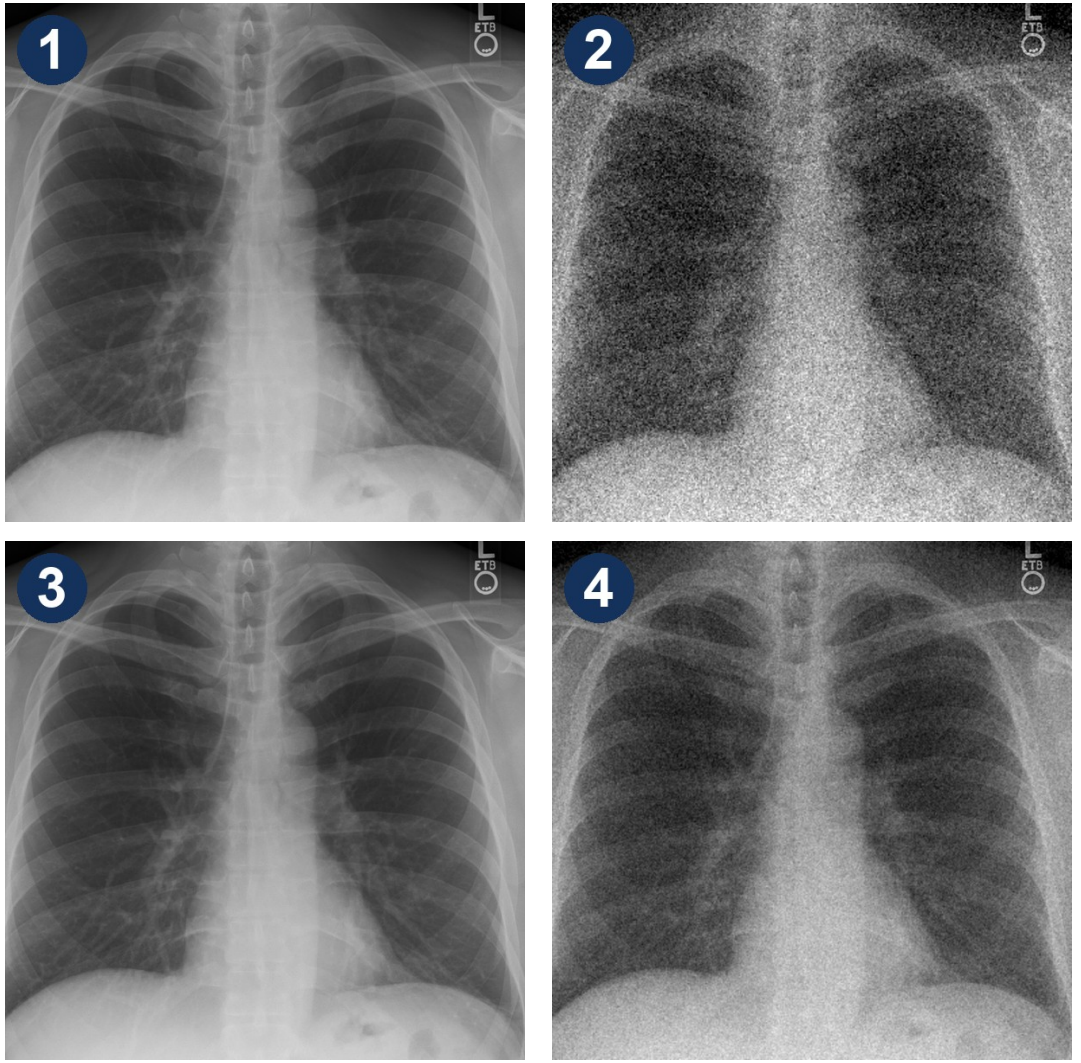
So how good are your eyes at judging exposure?

Redrawn from Seeram (2019) Figs 1.4 and 1.11, §1.6.2. Dynamic range: Seeram §1.6.1 (see Knowledge Guide for caveat).

Activity 1: the Radiologist Eye Test

ACTIVITY 1 · 1 min

- Four chest X-rays: same patient, different dose.
- Hold up 1–4 fingers: which got the LOWEST dose?
- Now: which got the HIGHEST?



Open DR_Lab.html → tab 2 “Warm-up”

Images: simulated by the group from a CC0 chest radiograph (Wikimedia Commons); noise exaggerated for projection. Run from DR_Lab.html, tab 2.

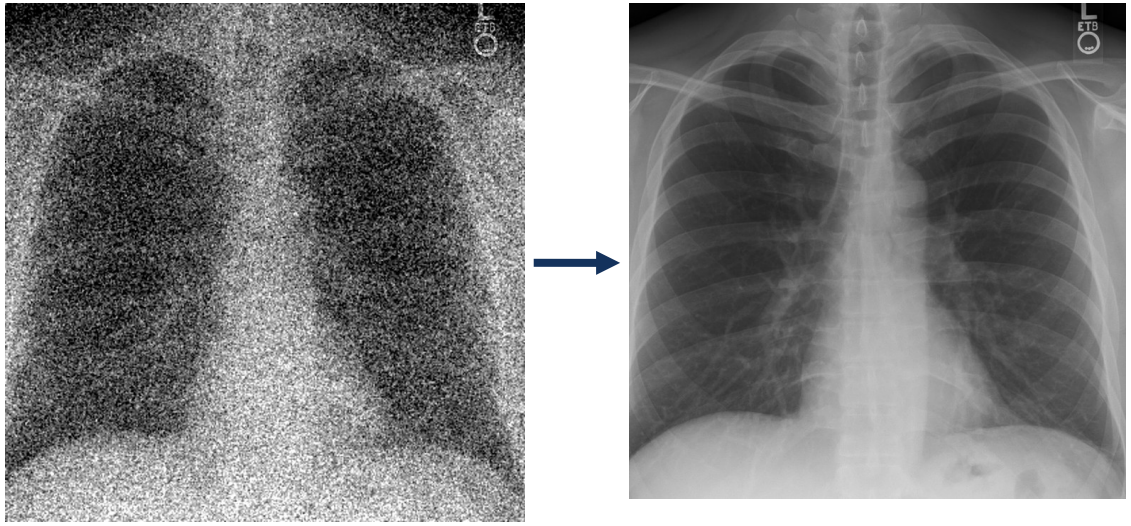
Activity 2: X-ray Blackjack

ACTIVITY 2 · 3 min

30 s per turn · 3:00 total

1. Both teams start from the same grainy image.
2. Each turn: play a × card. It multiplies your last scan, so cards chain.
3. Order: first team once, then the other twice in a row, then alternate (L R R L R L R L). No copying: not your opponent's last card.
4. 4 scans each, or STAND (timeout = STAND). Last image counts.

Score = Quality × 2 − Dose



Start: both teams → Ideal

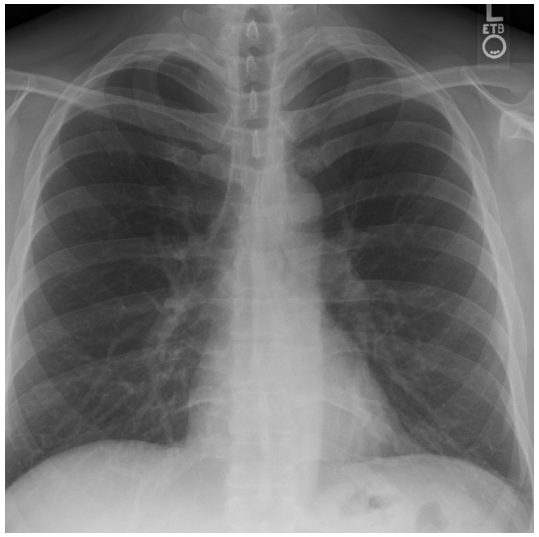
×⅙	×¼	×⅓	×½	×1.5	×2	×3	×4	×6

Cards chain on your last scan

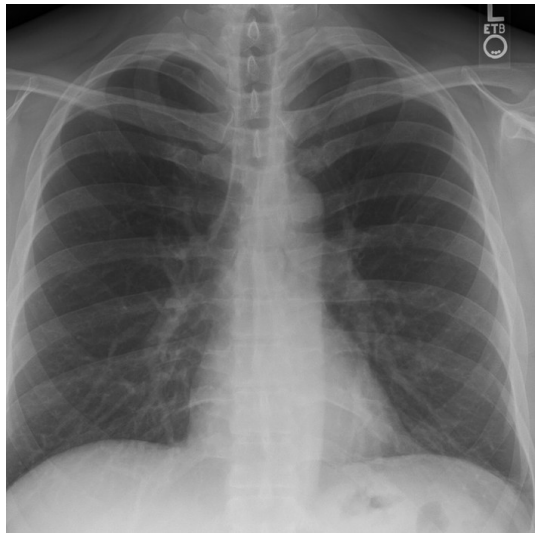
start	**1**
play ×6	**= 6 × start**
then ×1.5	**= 9 × start**

Images: simulated by the group from a CC0 chest radiograph (Wikimedia Commons); simplified model: noise ∝ 1/√photons, dose ∝ mAs (Seeram §1.3.5-1.3.6). DR_Lab.html, tab 3.

The number your eyes can't see: EI, DI and dose creep



1× · DI 0.0



3× · DI +4.8

Both look right. Only the DI shows who gave 3× the exposure.

$$DI = 10 \cdot \log_{10}(EI / EI_t)$$

EI: standardized detector-exposure number (not patient dose).
EI_t: target for the exam. +3 ≈ 2× exposure · −3 ≈ ½×



Dose creep: exposures drift upward because over-exposed digital images look fine.

ALARA: read the number, not just the image

AAPM Report No. 116 (2009) and IEC 62494-1 (2008); DI action bands as summarised in Seibert, IUPESM slides (see sources.md). Images: simulated by the group.

Beyond the chest X-ray: fluoroscopy and mammography

Digital fluoroscopy and DSA

Pre-contrast image − **Post-contrast image** = **Vessels only**

- Real-time moving images (GI tract, vessels)
- Image intensifier + TV/CCD → ADC
- Subtraction: temporal (in time) or energy (two kVp)

Digital mammography

CR | **Direct FPD** | **Indirect FPD** | **CCD**

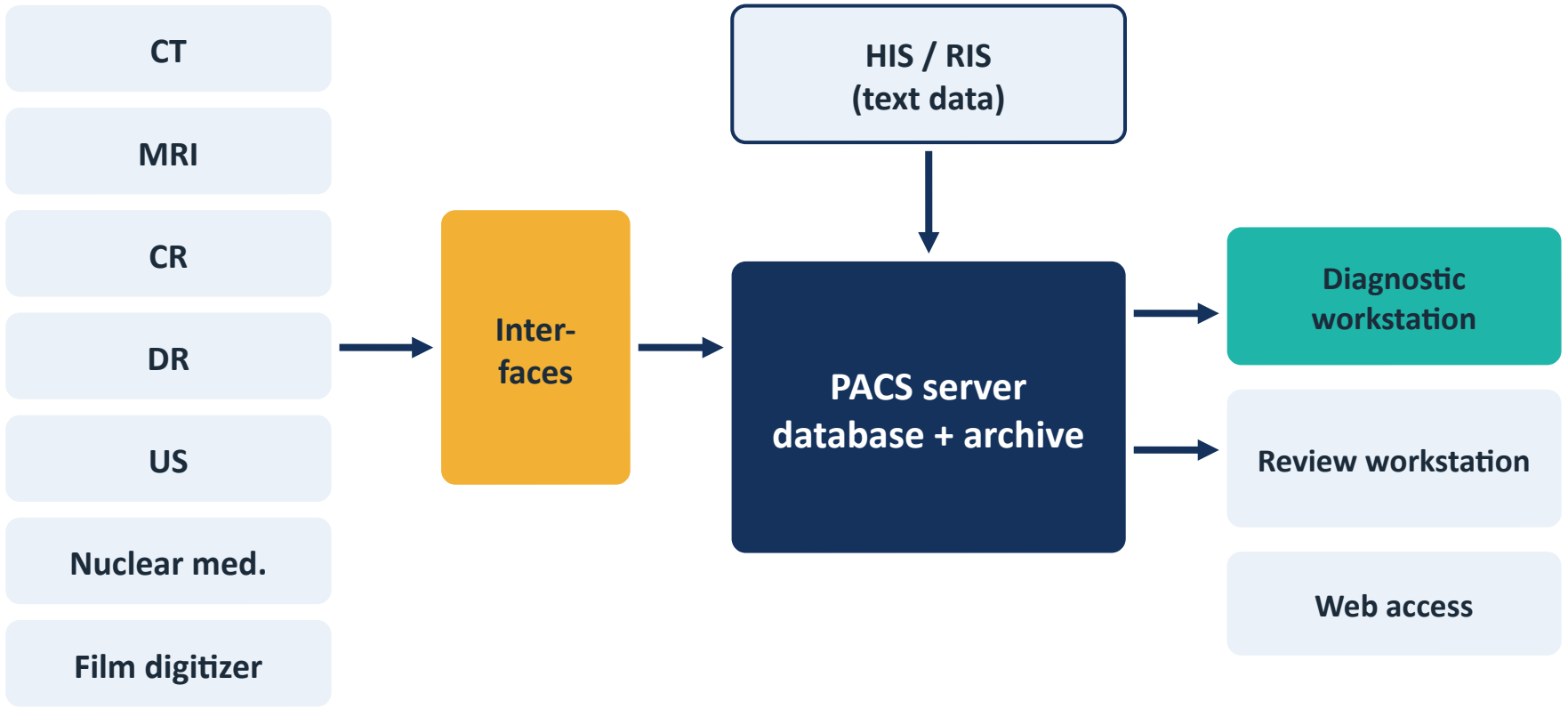
detectors used

- Digital post-processing to enhance interpretation
- CAD = “second reader” for microcalcifications and masses
- Leads to digital tomosynthesis

Digital tools: process, subtract and flag, after the exposure

Seeram (2019) §1.6.3-1.6.4, Figs 1.12-1.13. Diagram redrawn by the group.

PACS: the department's backbone



DICOM
images + metadata

HL7
text / patient information

IHE
makes them work together

Lossless (reversible) | **Lossy (irreversible)**

Compression

Next era: enterprise imaging · cloud · Big Data (volume, variety, velocity, veracity) · machine / deep learning · AI

Redrawn from Seeram (2019) Fig 1.14, §1.4.4-1.7, §1.9. IHE: RSNA + HIMSS, 1998. Next-era terms: Seeram §1.9.2.

Quality assurance and take-home

1

Film = detector + display + archive in one, with narrow latitude.

2

DR splits them: detector (CR / indirect / direct) → computer → display → PACS.

3

Wide latitude hides overexposure: read the EI / DI, not just the image.

QA / QC aims

- Minimum dose (ALARA)
- Optimum image quality for diagnosis
- Lower the cost of radiology operations

The technologist of the future is also an informaticist

Seeram (2019) §1.8 and §1.9.3; EI/DI: AAPM Report 116 (2009).

References and AI declaration

- Seeram E. Digital Radiography: Physical Principles and Quality Control. 2nd ed. Springer; 2019. Ch. 1. doi:10.1007/978-981-13-3244-9_1 (primary source)
- Seibert JA. Dose metrics: exposure index (summary of AAPM TG-116 / IEC 62494-1). IUPESM teaching slides, 2015.
- Leblans P, et al. Storage phosphors for medical imaging. Materials 2011;4:1034-86 (PMC5448636).
- AAPM Report No. 116. An exposure indicator for digital radiography. 2009.
- IEC 62494-1:2008. Exposure index of digital X-ray imaging systems.
- Chest radiograph: "Chest Xray PA 3-8-2010.png", Stillwaterising, Wikimedia Commons, CC0 1.0. All exposure/film simulations were generated by the group from it.

AI declaration: Claude (Anthropic) assisted with planning, game design, scripting and slide drafting. The group checked all content against Seeram (2019) and the references above, and re-derived the game scores by simulation.